Colouring minimal Cayley graphs: the road from Babai's conjectures to their resolution

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Abstract

In 1978 Babai conjectured that minimal Cayley graphs have a bounded chromatic number. In 1995 he changed his mind and conjectured the opposite.

1 AI/Data usage

2 Introduction

3 Preliminaries

3.1 Standard notation for paper

3.2 Cayley graphs with small examples

Definition 3.2.1 (Cayley Graphs)**.** Let G be a group, and let $S \subseteq G$ be a generating set, construct the graph $C(G, S)$ as follows:

- every $g \in G$ is a vertex,
- for every ordered pair (g, s), where $s \cdot g_1 = g_2$ with $s \in S$ and $g_1, g_2 \in G$, set an edge between g_1 and g_2, disallowing identical edges.

The resulting object, $C(G, S)$, is then called a Cayley Graph.

3.3 Minimal generating sets

3.4 Hypergraph coloring

4 Babai's question, 1978-1995

4.1 Minimal Cayley graphs are no-lonely-color

Definition 4.1.1 (Quasi-irreducible generating set)**.** Let G be a group, and let $S \subseteq G$ be a generating set. We call S *quasi-irreducible* if the group generated by S equals G, e.g.

$$\langle S \rangle = G,$$

and S can be well-ordered such that no element of S is contained in the group generated by the previous elements.

Example. Let $S = \{2, 3, 4\}$, then $3 - 2 = 1$. Since $\langle 1 \rangle = \mathbb{Z}$, also $\langle S \rangle = \mathbb{Z}$. Now observe that the the ordening $\{2, 4, 3\}$ fails since $4 \in \langle 2 \rangle$, but $\{4, 2, 3\}$ works since $2 \notin \langle 4 \rangle$ and $3 \notin \langle 4, 2 \rangle$. Hence S is a quasi-irreducible generating set of $\mathbb{Z}$.

Counterexample.

4.2 Forbidden subgraphs

4.3 Babai's 1978 conjectures

Both conjecture 2.2 and 3.5 from Babai's 1978 paper

4.4 The 1995 counter-conjecture

5 Evidence in both directions

5.1 Abelian groups are 3-colorable

5.2 More positive results

5.3 Negative results

6 Necessary tools

7 No-lonely-color graphs of large chromatic number

7.1 Tranquil hypergraphs

7.2 Tranquility of C_d

7.3 The construction

7.4 The proof

8 Minimal Cayley graphs of large chromatic number

9 Open questions

10 Appendix

References